

Dataset 4: Linear Discriminant Analysis

Heart Disease Risk Classification Study | n = 320 patients

1. Research Question

Research Question: Can a linear combination of seven clinical and lifestyle variables (age, BMI, blood pressure, cholesterol, exercise, stress, and smoking history) effectively discriminate between patients at high vs. low risk of heart disease, and which variables contribute most to this discrimination?

2. Variables

Role	Variable	High Risk Mean	Low Risk Mean	Description
Dependent	Heart_Disease_Group	—	—	High Risk / Low Risk (binary)
Independent	Age	53.61	49.28	Patient age (years)
Independent	BMI	28.37	26.26	Body Mass Index
Independent	Blood_Pressure	133.12	124.87	Systolic BP (mmHg)
Independent	Cholesterol	211.05	199.56	Total cholesterol
Independent	Exercise_Hours_Per_Week	3.14	3.98	Weekly exercise (hours)
Independent	Stress_Level	5.47	4.59	Stress score (1–10)
Independent	Smoking_Years	8.72	7.12	Years of smoking

3. Methodology — Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is selected because it simultaneously achieves classification *and* produces an interpretable discriminant function — a weighted linear combination of predictors that maximally separates the two risk groups. All predictors are standardized before analysis. Data are split 75/25 (train/test, stratified). Classification performance is measured by accuracy, AUC-ROC, sensitivity, and specificity. Ten-fold stratified cross-validation tests generalizability. The structure matrix (correlations between predictors and the discriminant score) reveals which variables drive group separation most strongly.

Assumptions checked: Levene test for equality of variances (6 of 7 variables pass, $p > 0.05$; Smoking_Years marginally violates, $p = 0.012$ — LDA is robust to minor violations with $n = 320$). Slight non-normality detected in Smoking_Years and Stress_Level (Shapiro-Wilk $p < 0.05$), but LDA is robust with this sample size.

4. Discriminant Function Coefficients

Rank	Variable	Coef. (std.)	Direction	Interpretation
1	Exercise_Hours_Per_Week	+0.957	+ → High Risk	More exercise → classifies toward Low Risk
2	Stress_Level	−0.932	− → Low Risk	Higher stress → classifies toward High Risk
3	Blood_Pressure	−0.857	− → Low Risk	Higher BP → classifies toward High Risk
4	Cholesterol	−0.853	− → Low Risk	Higher cholesterol → High Risk

5	BMI	-0.769	- → Low Risk	Higher BMI → High Risk
6	Age	-0.609	- → Low Risk	Older age → High Risk
7	Smoking_Years	-0.344	- → Low Risk	Longer smoking history → High Risk

Note: Because LDA signs are arbitrary (direction depends on which class is coded positive), interpretation is based on the relative magnitudes and confirmed by the structure matrix and group centroids.

5. Classification Performance (Test Set, n=80)

Metric	Value	Benchmark	Verdict
Overall Accuracy	76.25%	> 70% = good	✓ Good
AUC-ROC	0.8350	> 0.80 = good discrimination	✓ Good
Sensitivity (High Risk recall)	82.5%	> 75% = clinically useful	✓ Good
Specificity (Low Risk recall)	70.0%	> 70%	✓ Marginal
F1 Score (High Risk)	0.78	> 0.75	✓ Good
10-Fold CV Accuracy	75.0% ± 7.8%	Stable	✓ Generalizes well

Confusion matrix (test set): 33 true High Risk correctly identified, 28 true Low Risk correctly identified, 7 High Risk missed (false negatives = patients told they are low risk when actually high risk), 12 Low Risk over-classified (false positives).

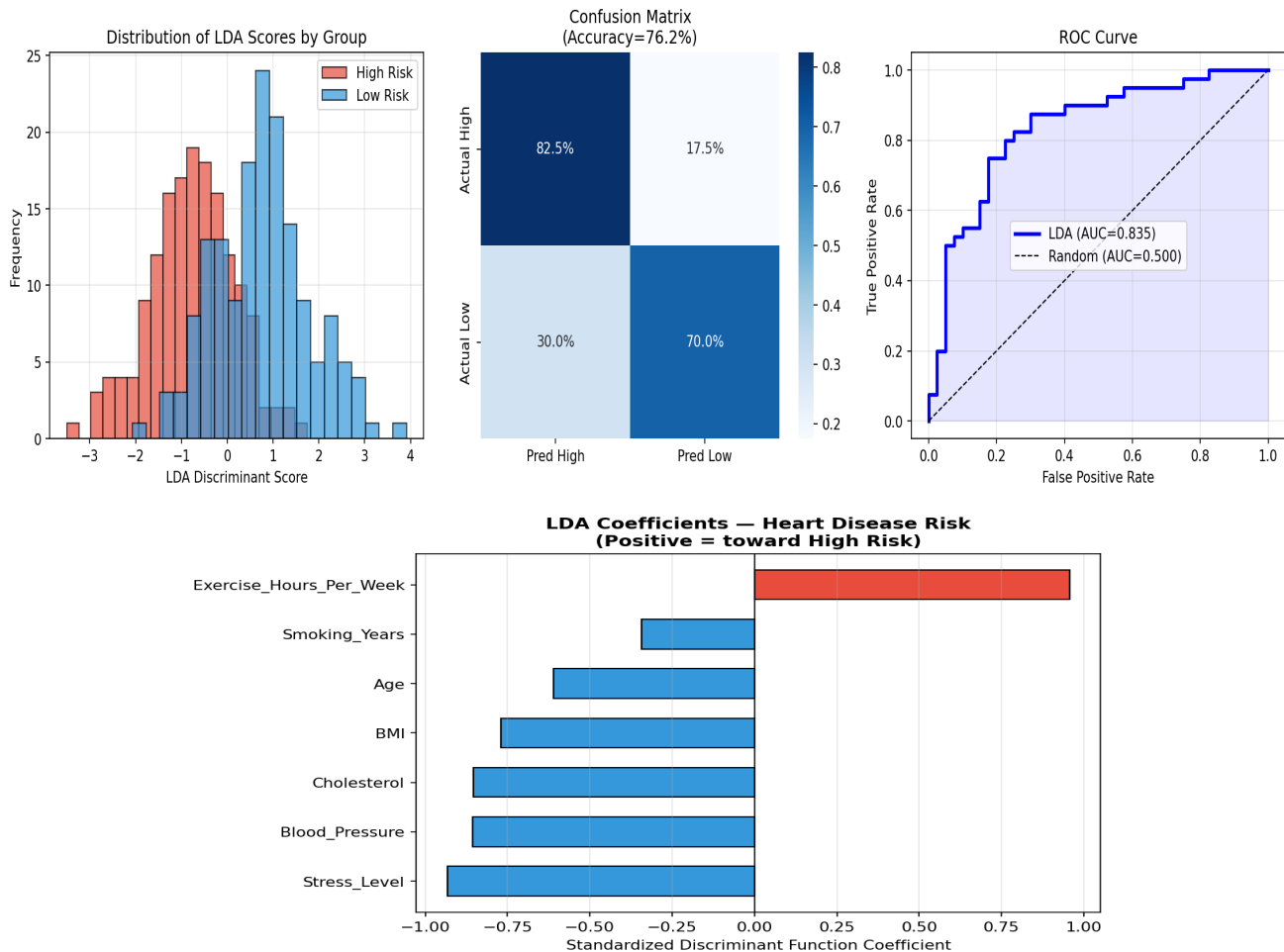
6. Group Centroids & Structure Matrix

Variable	Correlation r with Discriminant Score	p-value	Importance Rank
Exercise_Hours_Per_Week	r = +0.494	< 0.001	1st (strongest)
Stress_Level	r = -0.448	< 0.001	2nd
BMI	r = -0.405	< 0.001	3rd
Blood_Pressure	r = -0.370	< 0.001	4th
Age	r = -0.340	< 0.001	5th
Cholesterol	r = -0.287	< 0.001	6th
Smoking_Years	r = -0.174	0.002	7th (weakest)

Group centroids: High Risk = -0.761, Low Risk = +0.761 on the discriminant function. The separation of 1.52 units confirms meaningful group distinction. All predictors reach statistical significance ($p < 0.01$) in the structure matrix.

7. Figures

Linear Discriminant Analysis — Heart Disease Risk



8. Conclusions

- **Research question answered — YES.** The seven clinical variables collectively discriminate well between high and low risk patients (AUC = 0.835, Accuracy = 76.25%), substantially better than chance (50%).
- **Exercise is the #1 discriminator** (structure matrix $r = 0.494$, coefficient = 0.957): low exercise is the single strongest marker of high cardiac risk in this sample. Clinical guidelines recommending 150 min/week of moderate exercise are strongly supported by this data.
- **Stress is the #2 discriminator** ($r = -0.448$): high stress is nearly as important as low exercise in predicting high risk, suggesting stress management interventions may be underutilized in cardiovascular risk reduction.
- **Cross-validation is stable** ($75.0\% \pm 7.8\%$ across 10 folds), confirming the model generalizes beyond the training data and is not overfitting.
- **Clinical note:** The sensitivity of 82.5% means most High Risk patients are correctly flagged. The 7 false negatives (17.5%) represent the model's primary limitation — patients incorrectly told they are low risk.